ASVD-Bridge: Zero-Overhead Subspace Stabilization for Deep Low-Rank Transformer Compression

Trentzap
https://github.com/Trentzap1/qtensor-engine
https://huggingface.co/trentzap/QTensor-Llama-3-8B-Demo

August 18, 2026

Abstract

The phase transition collapse of singular values poses a fundamental barrier to deeply compressing Large Language Models (LLMs) via low-rank matrix decomposition. When multi-head attention projections are aggressively truncated, the activation variance across the residual stream plummets, inducing severe representational degradation in downstream transformer blocks. We introduce **ASVD-Bridge** (*Asymmetric Singular Value Decomposition with Subspace Stabilization*), an information-theoretic hybrid compression framework. ASVD-Bridge pairs continuous dynamic SVD rank allocation in attention mechanisms with discrete 4-bit (INT4) quantization across Multilayer Perceptron (MLP) blocks. To counteract initial truncation shock during Quantization-Aware Distillation (QAD), we insert a learnable Rank-1 Subspace Bridge (γ) at the projection boundaries. Crucially, during artifact compilation, γ and the low-rank adaptation parameters fold deterministically into the static projection matrices, guaranteeing **zero latency and zero FLOP overhead** at inference time. We validate our framework on the `Meta-Llama-3-8B` architecture, compressing the model to a **4.67 GB physical VRAM footprint** while achieving **32.93 Tokens Per Second (TPS)** on consumer-grade hardware.

1 Introduction

As Large Language Models (LLMs) scale in parameter depth and width, deploying high-capacity models on edge devices remains bottlenecked by memory bandwidth and VRAM constraints. Post-training quantization (PTQ) and low-rank approximation (e.g., Singular Value Decomposition) have emerged as primary paradigms for model compression. However, applying unconstrained low-rank truncation directly to deep attention layers fundamentally deforms the geometry of the transformer's residual stream.

To conceptualize this failure mode, consider a deep neural network as an interconnected municipal water grid. Mathematically compressing a weight matrix shrinks the diameter of the transmission channels. While principal singular vectors preserve the dominant directional flow, the overall activation variance (the mathematical "pressure") drops precipitously. When this attenuated signal enters subsequent non-linear activation layers, the severe variance mismatch destabilizes downstream feature extraction, precipitating catastrophic forgetting.

The **Subspace Bridge** acts as a dynamic pressure regulator positioned directly at the truncated projection boundaries. By introducing a single learnable scalar during distillation, the network smoothly aligns the low-rank activation manifold with the teacher's original variance profile. Furthermore, because this stabilization operates as a Rank-1 affine transformation, it is mathematically fused into the adjacent projection weights post-training. This entirely eliminates the module at runtime, yielding zero latency overhead during local execution.

2 Related Work

Model compression methodologies broadly divide into uniform discrete quantization and continuous low-rank decomposition:

Uniform Discrete Quantization: Methods such as AWQ, GPTQ, and GGUF cluster continuous floating-point weights into discrete 4-bit bins. While effective at preserving perplexity at moderate compression ratios, discrete clustering restricts continuous gradient adaptation and introduces non-trivial runtime dequantization overhead on consumer hardware.

Low-Rank Factorization: Structural methods, including SVD-LLM and SliceGPT, factorize dense weight matrices into lower-rank components. While computationally continuous, aggressive rank truncation inevitably causes activation variance collapse across deep architectures, typically requiring expensive second-order Hessian corrections.

Variance Calibration: Learnable residual scaling has been explored in training deep architectures, notably by LayerScale and ReZero. Unlike permanent architectural layers, our Subspace Bridge serves as a transient optimization manifold that dissolves completely upon weight export.

3 Methodology

3.1 Information-Theoretic Asymmetric Partitioning

Transformer blocks exhibit functional dichotomy: Multi-Head Attention projections (W_q, W_k, W_v, W_o) execute continuous routing over spatial token dependencies, whereas MLP blocks ($W_{gate}, W_{up}, W_{down}$) act as high-capacity key-value associative memories.

Because attention manifolds are hypersensitive to quantization noise, we enforce an *asymmetric compression topology*:

1. **Feed-Forward Blocks (MLP):** Quantized to physical 4-bit representations (INT4) with output channel slicing to maximize parameter density.
2. **Attention Blocks:** Factorized via continuous Singular Value Decomposition into rank-r factors:
$$W \approx U_r \Sigma_r V_r^T = A \cdot B \tag{1}$$
where $A \in \mathbb{R}^{d_{in} \times r}$ and $B \in \mathbb{R}^{r \times d_{out}}$, with the rank r dynamically allocated based on layer-wise Shannon entropy.

3.2 The Rank-1 Subspace Bridge

For an input activation $X \in \mathbb{R}^{B \times S \times d_{in}}$, passing through truncated factors A and B scales the empirical activation variance by a factor proportional to $\sum_{i=1}^{r} \sigma_i^2 / \sum_{j=1}^{d} \sigma_j^2 < 1$.

To isolate this compression shock during training, we attach a learnable scalar $\gamma \in \mathbb{R}$ initialized to calibrate output magnitude:
$$\hat{X} = \gamma \cdot ((XA)B) \tag{2}$$

During Quantization-Aware Distillation (QAD), γ absorbs the bulk variance recovery gradients, permitting the paired Low-Rank Adaptation (LoRA) modules to optimize fine-grained semantic tokens rather than bulk activation scaling.

3.3 Zero-Overhead Deterministic Weight Folding

Following distillation convergence, the temporary optimization graph is permanently collapsed. Let $A_{base} \in \mathbb{R}^{d_{in} \times r}$ and $B_{base} \in \mathbb{R}^{r \times d_{out}}$ denote the frozen SVD matrices, and let $\Delta A = A_{down} A_{up}$ and $\Delta B = B_{down} B_{up}$ denote the trained LoRA adapters scaled by $\frac{\alpha}{r_{lora}}$.

The final deployment matrices A_{final} and B_{final} are computed via:

$$A_{final} = A_{base} + \frac{\alpha}{r_{lora}} (A_{down} A_{up}) \tag{3}$$

$$B_{final} = \gamma \cdot \left(B_{base} + \frac{\alpha}{r_{lora}} (B_{down} B_{up}) \right) \tag{4}$$

Because scalar multiplication is associative over matrix products:

$$\hat{X} = \gamma \cdot (X A_{final}) B'_{final} = (X A_{final}) B_{final} \tag{5}$$

where $B_{final} = \gamma \cdot B'_{final}$. The compiled model runs entirely through standard GEMM kernels with zero auxiliary parameters and zero additional FLOPs.

3.4 Quantization-Aware Distillation (QAD)

To re-anchor the student representations to the uncompressed teacher manifold, we minimize a masked Kullback-Leibler (KL) divergence objective:

$$\mathcal{L}_{QAD} = \frac{1}{\sum M_{i,j}} \sum_{i,j} M_{i,j} \cdot \mathcal{D}_{KL} \left(P_{teacher}(y_{i,j} \mid x) \parallel P_{student}(y_{i,j} \mid x) \right) \tag{6}$$

where $M_{i,j} \in \{0, 1\}$ is a binary mask excluding all <pad> tokens. This masking prevents gradient dissipation over padding tokens, forcing the optimizer to focus strictly on semantic recovery.

4 Experiments and Hardware Results

We evaluated the ASVD-Bridge architecture on `NousResearch/Meta-Llama-3-8B`. Benchmarking was conducted on a single consumer NVIDIA GeForce RTX 5080 GPU (16 GB GDDR7, CUDA 13.2).

Table 1: Inference and Memory Telemetry: FP16 Baseline vs. ASVD-Bridge (Step 230 Alpha)

Model Configuration	**VRAM Footprint**	**Throughput**	**QAD Loss**	**Coherence Sta**
Meta-Llama-3-8B (FP16 Baseline)	∼15.0 GB	15.12 t/s	—	Full Pre-traine
ASVD-Bridge 8B (Step 230 Demo)	**4.67 GB**	**32.93 t/s**	**5.2695**	Early Syntax He
ASVD-Bridge 1B (Proof-of-Concept)	0.94 GB	78.40 t/s	2.1840	Fully Converg

Observation on Step 230 Checkpoint: The Step 230 checkpoint demonstrates that the ASVD-Bridge stabilizes the structural forward pass, reducing memory from 15.0 GB to 4.67 GB while doubling throughput (32.93 TPS). As expected during early distillation ($\mathcal{L} \approx 5.26$), token boundaries and grammatical clauses have re-formed, while deep semantic alignment continues to converge toward the 5,000-step target.

5 Conclusion and Limitations

The ASVD-Bridge demonstrates that high-parameter transformer architectures can execute within sub-5 GB consumer memory bounds via asymmetric low-rank factorization and folded subspace stabilization.

Limitations: The current public demo artifact reflects an early distillation checkpoint (Step 230) released to validate hardware throughput and memory limits; factual generation remains uncalibrated until full distillation plateau completion. Additionally, distillation is currently bounded by the Alpaca instruction-tuning distribution (52K samples). Future iterations will extend distillation across broader multi-turn synthetic datasets.